

Computer Science 1MD3 -- Introduction to Programming

ADMINISTRATIVE DETAILS

Instructor: Dr. Vincent Maccio

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TOPICS AND LEARNING OBJECTIVES

Calendar Description: Introduction to fundamental programming concepts: values and types, expressions and evaluation, control flow constructs and exceptions, recursion, input/output and file processing.

Learning Objectives: This course focuses on small program/function implementation, starting with the absolute basics of programming and working up to recursion and other applications.

By the end of this course students will be able to:

- Write functions in Python
- Trace and implement simple programs
- Understand variable scope and types
- Understand control flow structures including:
 - If statements, nested if statements
 - For loops, while loops, nested loops
- Use (nested) lists and (nested) dictionaries to implement Python functions
- Read and write to files using Python
- Implement recursive functions

MATERIALS & FEES

There is no required text for this course. **Total cost: \$0.00**

I will post weekly practice problems on Avenue.

COURSE OVERVIEW AND ASSESSMENT

All course material, announcements, assignments, etc., will be posted on Avenue. During lectures I typically use a mixture of live coding and the black/whiteboard. I intend to record my lectures. However, I make no guarantee I will do so. That is, technical issues occur; things out of my control happen. If a lecture goes unrecorded, it is your responsibility to get caught up. There are two separate lecture sections in this course.

Assessments:

Item	Weight	Comments
Assignment 0 (week 2)	1%	Just to make sure you understand auto-grading
Assignment 1 (week 5)	4%	Python basics: Booleans, strings, if statements
Assignment 2 (week 9)	5%	Nested loops, nested lists
Midterm (Nov. 3rd)	25%	Held during the evening. The lecture on Oct 30th will be a review session, there will be no lecture on Nov 3rd.
Assignment 3 (week 12)	5%	Recursion, file I/O, dictionaries
Exam (Exam week)	45%	Cumulative.
Test Floating	10%	Better midterm or exam grade
Assignment Floating	5%	Best assignment grade between Assignment 1, 2, and 3

The due dates above are rough estimations. **I will always give you at least two weeks to complete an assignment from the date I post it.** There will be a grace period of 5 days where you can submit late without penalty. **No further individual extensions will be granted;** if the class is struggling and this is brought to my attention early enough, I may grant an extension for the entire class. The assignment “topics” are also rough estimations. Late assignment submissions after the grace period will receive a grade of 0.

All missed work with valid justification (for example, MSAF) will have its weight be reallocated to the final exam. There will be no “makeup assignments”.

LECTURES

This course has three 1-hour lectures a week and one 1-hour tutorial each week. I intend to record and livestream all my lectures but make no guarantee that I will do so. These recordings are not meant as a one-to-one replacement for attending lectures, but rather an aid to get caught up if you miss a lecture. It is ultimately your responsibility to keep up with the course material.

TUTORIALS

<u>Section</u>	<u>TA</u>
T01	Muneeb
T02	Aarav
T03	Oroni
T04	Praveena
T05	Jiajun
T06	Aidan
T07	Aarav
T08	TBD
T09	Oroni

ACADEMIC INTEGRITY

You are expected to exhibit honesty and use ethical behaviour in all aspects of the learning process. Academic credentials you earn are rooted in principles of honesty and academic integrity. **It is your responsibility to understand what constitutes academic dishonesty.**

Academic dishonesty is to knowingly act or fail to act in a way that results or could result in unearned academic credit or advantage. This behaviour can result in serious consequences, e.g. the grade of zero on an assignment, loss of credit with a notation on the transcript (notation reads: “Grade of F assigned for academic dishonesty”), and/or suspension or expulsion from the university. For information on the various types of academic dishonesty please refer to the Academic Integrity Policy, located at <https://secretariat.mcmaster.ca/university-policies-proceduresguidelines>

The following illustrates only three forms of academic dishonesty:

- plagiarism, e.g. the submission of work that is not one’s own or for which other credit has been obtained.
- improper collaboration in group work.
- copying or using unauthorized aids in tests and examinations.

The Canadian Engineering Accreditation Board (CEAB) is a division of Engineers Canada and is responsible for accrediting undergraduate engineering programs across Canada. Accreditation by the CEAB ensures that the engineering programs meet a national standard of quality and cover essential educational requirements. Graduate Attributes are a set of qualities and skills that the CEAB expects engineering graduates to possess. These attributes are a benchmark for the learning outcomes of accredited engineering programs. Since graduate attributes are broad and abstract, McMaster’s Faculty of Engineering has refined them into more specific indicators. This section lists the Graduate Attribute Indicators associated with the Learning Outcomes in this course.

- Graduate Attribute Indicator: Knowledge (all learning outcomes are associated with this Indicator)

AUTHENTICITY / PLAGIARISM DETECTION

Some courses *may* use a web-based service (Turnitin.com) to reveal authenticity and ownership of student submitted work. For courses using such software, students will be expected to submit their work electronically either directly to Turnitin.com or via an online learning platform (e.g. A2L, etc.) using plagiarism detection (a service supported by Turnitin.com) so it can be checked for academic dishonesty.

Students who do not wish their work to be submitted through the plagiarism detection software must inform the Instructor before the assignment is due. No penalty will be assigned to a student who does not submit work to the plagiarism detection software. **All submitted work is subject to normal verification that standards of academic integrity have been upheld** (e.g., on-line search, other software, etc.). For more details about McMaster's use of Turnitin.com please go to www.mcmaster.ca/academicintegrity.

COURSES WITH AN ON-LINE ELEMENT

Some courses *may* use on-line elements (e.g. e-mail, Avenue to Learn (A2L), LearnLink, web pages, capa, Moodle, ThinkingCap, etc.). Students should be aware that, when they access the electronic components of a course using these elements, private information such as first and last names, user names for the McMaster e-mail accounts, and program affiliation may become apparent to all other students in the same course. The available information is dependent on the technology used. Continuation in a course that uses on-line elements will be deemed consent to this disclosure. If you have any questions or concerns about such disclosure please discuss this with the course instructor.

ONLINE PROCTORING

Some courses *may* use online proctoring software for tests and exams. This software may require students to turn on their video camera, present identification, monitor and record their computer activities, and/or lock/restrict their browser or other applications/software during tests or exams. This software may be required to be installed before the test/exam begins.

CONDUCT EXPECTATIONS

As a McMaster student, you have the right to experience, and the responsibility to demonstrate, respectful and dignified interactions within all of our living, learning and working communities. These expectations are described in the Code of Student Rights & Responsibilities (the "Code"). All students share the responsibility of maintaining a positive environment for the academic and personal growth of all McMaster community members, **whether in person or online**.

It is essential that students be mindful of their interactions online, as the Code remains in effect in virtual learning environments. The Code applies to any interactions that adversely affect, disrupt, or interfere with reasonable participation in University activities. Student disruptions or behaviours that interfere with university functions on online platforms (e.g. use of Avenue 2 Learn, WebEx or Zoom for

delivery), will be taken very seriously and will be investigated. Outcomes may include restriction or removal of the involved students' access to these platforms.

ACADEMIC ACCOMMODATION OF STUDENTS WITH DISABILITIES

Students with disabilities who require academic accommodation must contact Student Accessibility Services (SAS) at 905-525-9140 ext. 28652 or sas@mcmaster.ca to make arrangements with a Program Coordinator. For further information, consult McMaster University's Academic Accommodation of Students with Disabilities policy.

REQUESTS FOR RELIEF FOR MISSED ACADEMIC TERM WORK

McMaster Student Absence Form (MSAF): In the event of an absence for medical or other reasons, students should review and follow the Academic Regulation in the Undergraduate Calendar "Requests for Relief for Missed Academic Term Work".

All missed work will have their weight shifted to the final exam.

ACADEMIC ACCOMMODATION FOR RELIGIOUS, INDIGENOUS OR SPIRITUAL OBSERVANCES (RISO)

Students requiring academic accommodation based on religious, indigenous or spiritual observances should follow the procedures set out in the RISO policy. Students should submit their request to their Faculty Office **normally within 10 working days** of the beginning of term in which they anticipate a need for accommodation or to the Registrar's Office prior to their examinations. Students should also contact their instructors as soon as possible to make alternative arrangements for classes, assignments, and tests.

COPYRIGHT AND RECORDING

Students are advised that lectures, demonstrations, performances, and any other course material provided by an instructor include copyright protected works. The Copyright Act and copyright law protect every original literary, dramatic, musical and artistic work, **including lectures** by University instructors.

The recording of lectures, tutorials, or other methods of instruction may occur during a course. Recording may be done by either the instructor for the purpose of authorized distribution, or by a student for the purpose of personal study. Students should be aware that their voice and/or image may be recorded by others during the class. Please speak with the instructor if this is a concern for you.

EXTREME CIRCUMSTANCES

The University reserves the right to change the dates and deadlines for any or all courses in extreme circumstances (e.g., severe weather, labour disruptions, etc.). Changes will be communicated through regular McMaster communication channels, such as McMaster Daily News, A2L and/or McMaster email.